HANDBOOK | FOR FIRST-COURSE STUDENTS

The ShopBot PRSalpha

144-60

How it works, how to run it, what goes wrong. The printed companion to the lab's teaching site.

LAB	DATE	VERSION
Prototyping Lab, University of Iceland	2026-10-21	Handbook 2 · this machine: Alpha control Box Model# **10372**, firmware 224; SB3 version 3.8.70.

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References: Appendix C. Read chapters 01–05 before your induction, tear out the pre-flight sheet (page 10) and keep it at the machine.

Before you touch it

Ten rules and two procedures. Know them before the induction.

1 **Trained and signed off.**
You run this machine only after the induction and the instructor's sign-off. Until then, only with a signed-off person at the E-stop.

2 **Never alone, never tired.**
Two people in the room; the second one knows where the E-stop is: the red button on the wall-mounted pendant by the monitor. It is the only one.

3 **Eyes and ears, for you and anyone watching.**
From the moment the machine is powered. No gloves, loose clothing, hair or lanyards near the gantry.

4 **Hands out of the cutting area while the control box is on.**
The only exception: spindle stopped, keypad closed, no file loaded or paused, and the key OFF in your pocket. It can move without warning, in any direction.

5 **Key out for hands-on work.**
Spindle stopped, interlock key OFF and in your pocket before you change a bit, touch the collet, or load, move or clamp material.

6 **Never leave a running job.**
Dust under the brush catches when a tool rubs, hits a screw or burns hardwood. Stand where you can see the cut and reach the E-stop.

7 **Know the two ways to stop.**
Spacebar for routine; E-stop when something is wrong. Use them without hesitating; a ruined sheet is cheap.

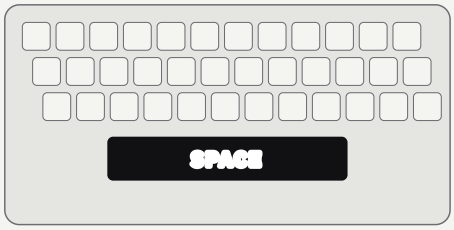
8 **Spindle at speed before OK.**
When the screen says “Press START on control before pressing OK”: pendant START, check the spindle runs at the commanded speed, then OK — cutting begins. No separate RUN on the VFD. A still bit dragged through plywood snaps.

9 **Extraction on.**
Switch the dust collector on at the collector itself (manual; SB3 does not switch it). It runs whenever the spindle runs. If the hose comes off the foot or the brush skirt is damaged, stop and tell staff.

10 **Clean up; leave the machine powered.**
Stop the file and spindle; key out before cleanup. Clear and clean the deck, stow the bit, report faults. Users never power down the machine; leave it for lab staff. Use E-stop in an emergency, not to finish a session.

How to stop

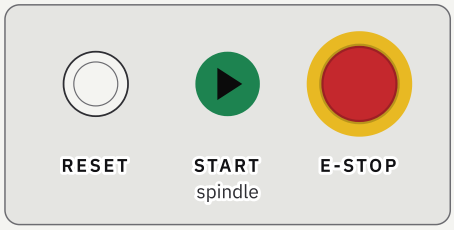
Two controls, two meanings. Use them without hesitating.



ROUTINE · POSITION KEPT

Spacebar

Or Enter, or on-screen Stop. Motion ramps down, Z lifts, position is kept, you can resume. **Treat the spindle as live** — it may keep turning, coast, or restart on resume. Before reaching in: spindle stopped, key OFF and removed, keypad closed, no file loaded or paused. Keep clear before restarting. [4,11]



SOMETHING IS WRONG · POSITION LOST

E-stop

The red button on the wall-mounted pendant by the monitor — the only E-stop on this machine (the red-and-yellow rotary handle on the control cabinet is the power disconnect, not an E-stop). Spindle power off at once, motors ~5 s later. Position is **lost**: twist to release → pendant **Reset** → `C3` → `C2`. If the software has frozen, E-stop is the only control that still kills the spindle. [2,10,11]

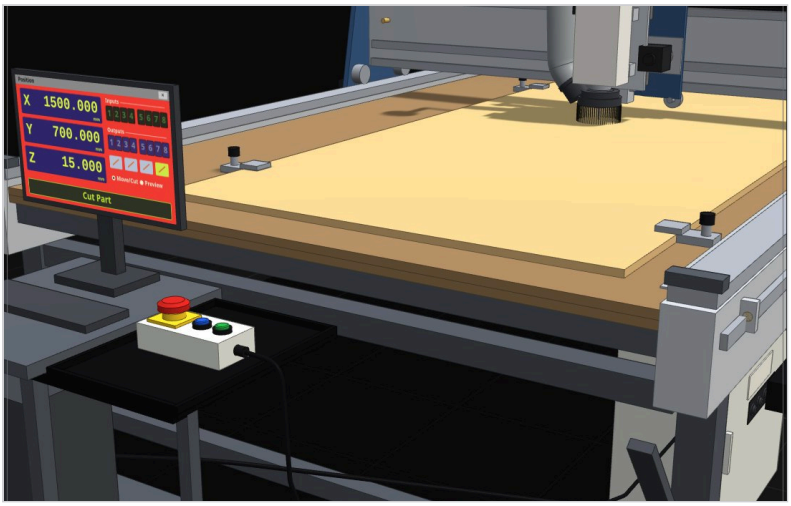
How to start a job

The one sequence. The pendant comes before OK.

1. **RPM typed, Enter**
   Spindle RPM Control window (Tools menu or `TR`); leave it open.
2. **`FP`, pick file, Start**
   Offset = No Offset.
3. **Pendant START**
   When the screen says “Press START on control before pressing OK”. Green button.
4. **Spindle at speed**
   Check it runs at the commanded rpm (the VFD display). There is no separate RUN step on the VFD.
5. **OK**
   Cutting begins. Hand near the spacebar for the first plunge.

What moves, and why

Four motors, three axes, one spindle. What each part does, and what the motors cannot see.



The machine in 3D. Still from the site's 3D model: table and spoilboard, the gantry (X bridge), the Y car carrying the spindle, the control PC and the wall pendant with the only E-stop. Online the model turns and acts out ten faults; five are frozen in chapter 06.

SHEET	12 × 5 ft
TRAVEL	153 × 61 in
MOTORS	4 closed-loop alphaStep
DRIVE	7.2 : 1, rack and pinion
RAPID	30 in/s · 760 mm/s
SPINDLE	12 000 rpm typical, 6–18k
RESOLUTION	0.0004 in ≈ 0.01 mm

Staff only, not part of the student course: the rotary-axis chuck standing beside the table belongs to this machine but is not used in student jobs. Leave it alone; ask staff.

HOW IT WORKS

How the mechanisms work

The gantry (X)

The bridge is an aluminium extrusion between two end plates. Each plate hangs four V-rollers on the hardened rail that runs along the top of the side plate; two ride above, two below, so the car cannot lift. A motor on each end plate sits on a hinged mount pulled by a spring (3.5 turns past contact on spring mounts; on bolted motor plates, hold the pinion into the rack with 15–20 lb while tightening the four plate screws) so its pinion stays pressed into the rack on the outside of the side plate. [2,13] Both X motors receive identical step trains. If one side stalls, its driver alarms and everything stops. If a pinion jumps teeth or slips on its shaft, the motor still turns the commanded amount: the bridge racks out of square *silently*.

The car (Y) and the Z plate

The YZ car repeats the same idea rotated 90°: rollers on a V-rail across the face of the beam, one motor on a pinion in the beam's rack. The Z plate rides one captured rail between two rows of five V-bearings (three fixed, two eccentric) per side and is lifted by a vertical rack — no leadscrew — so it moves at the same 7.2:1 ratio as the other axes. Because the gear train back-drives, Z can drop when unpowered; never push any axis by hand with the motors connected — position is lost and the drivers can be damaged. [2,18,27] Too tight on the Z eccentrics means binding and lost position; too loose means a lateral slop you will see as a bad fourth side on every pocket. [18]

Closed loop, but closed around the motor

An alphaStep motor runs open-loop until rotor position deviates by ±1.8° from command; at that point the driver switches to closed-loop correction. If correction cannot hold (an obstruction, a wrong feed, a damaged cable), the driver alarms, the control board stops everything, and the software says the location is no longer accurate. [1,22,23,30] Red LED blink codes on the AS driver: 1 overheat (~85 °C heatsink), 2 overload (>5 s above max torque), 3 overvoltage, 4 speed error, 5 overcurrent, 6

overspeed, 7 EEPROM, 8 sensor.[22] What the loop cannot see: a pinion slipping on its shaft, a bit creeping in the collet, or a sheet moving under the clamps.

Spindle and VFD

The VFD makes the spindle's three-phase supply; the control box only tells it *on* (output 1 through the key interlock and the pendant Start) and, on packages with the RPM controller, *how fast* (the `TR` window or `TR,12000` in a file, SB3.5.5 or newer).[4,21,53] The lab's full sequence: the screen prompts “Press START on control before pressing OK” → pendant Start → check the spindle runs at the commanded speed → OK → cutting begins. There is no separate RUN step on the VFD.[4] Warm-up (`C5`, the lab's standard routine): 9000, 10 000 and 12 000 rpm for 3 min each at the start of the day or after more than 4 h idle.[4,13] Chipload = feed ÷ (rpm × flutes); the figure runs a 6 mm two-flute bit at 60 mm/s and 12 000 rpm, 0.15 mm per tooth.[11]

Zeroing and homing are chapter 04; dust extraction and hold-down are chapters 01 and 03.

REFERENCE

The machine in numbers

“144-60” is the nominal sheet class, not the footprint. The 2023 specification sheet lists a **153 × 61 × 8 in (3.89 × 1.55 × 0.20 m) movement area** and a **168 × 91 × 67 in (4.27 × 2.31 × 1.70 m) footprint**; the May 2022 assembly guide still describes the conventional **6 in (152 mm) Z**, with 12 in and 24 in tall-Z options.[1,2] These are catalogue reference dimensions, not usable clearance for a job. Before running a tall setup, have staff check that the stock, fixtures, tool length and retract height fit. Do not disable limits.

table
bolted steel legs and supports with aluminium T-slot side extrusions; **not** a welded one-piece frame. Base layer 3/4 in plywood on 3/8 in carriage bolts, replaceable 3/4 in MDF spoilboard on top.[1,2,3]

rails
hardened steel **V-rails** with **V-roller** bearings on X and Y; eccentric rollers set the preload on the YZ car. Z rides a captured rail on five bearings per side (three concentric, two eccentric).[1,13,18,19]

drive
rack and pinion on all three axes, including Z. Two X motors (one per side), one Y, one Z. Pinions: 30 teeth on X and Y, 25 on Z. Gearboxes 7.2:1 low-backlash tapered-hob.[1,2,27]

motors
This machine uses Oriental Motor **ARM98AC-T7.2 alphaStep** closed-loop steppers. Feedback is at the motor shaft — not at the rack, the bit or the workpiece.[2,23,27]

resolution
2482.8 steps/in on X and Y (0.0004 in ≈ 0.010 mm), 2979.4 steps/in on Z, resolution multiple 5.[1,27]

speed
positioning 1800 in/min (30 in/s, 762 mm/s) XY and 900 in/min Z; up to 720 in/min (12 in/s) XY and 360 in/min Z under full cutting force; about 150 lbf of cutting force at 60 in/min.[1]

control
Windows PC → USB → PRSalpha control box (control board, four drivers, contactors) → motors. Files are OpenSBP `.sbp` text. 8 inputs / 8 outputs at 5 V logic (~5 mA). This machine's control box: Alpha control Box Model# **10372**, firmware 224; SB3 version 3.8.70.[1,2,10,11]

spindle
Spindle: **HSD AT/MT 1073-140**. Nameplate rating: 3 kW **S6**, 220 V three-phase, 300 Hz, 18,000 rpm, 13 A. S6 is an intermittent periodic duty rating, not an S1 continuous-power claim, and 18,000 rpm is the nameplate operating speed, not a confirmed maximum. Spindle controller: **Yaskawa V1000, CIMR-VU2A0020GAA**. **ER25** collets with the standard ER25 nut supplied for this HSD spindle model. 24 V cooling fan. (ShopBot's packages: HSD 2.2 or 4 HP, or Colombo 5 HP.)[2,20,21,41]

sensors
input 1 Z-zero plate, input 2 X proximity, input 3 Y proximity, input 4 stop circuit, input 5 Z proximity; output 1 spindle, output 4 safety/spindle-enable.[2]

Your first job

Eleven steps from CAM to a finished part, with the command at each step.

AT THE COMPUTER	1	**CAM** Mostly Autodesk Fusion. Stock measured, origin corner, Z0 surface, screws and clamps clear of every toolpath, tabs. Post with the ShopBot OpenSBP post, mm.	`Fusion · OpenSBP · mm`
	2	**Read the .sbp** `&PWZorigin`, units, `MS`, `TR`, first `JZ` / `M3`.	`&PWZorigin` `MS` `TR` `JZ` `M3`
POWER AND HOME	3	**Power** Control box on → E-stop twisted to release, pendant Reset → PC on, open SB3 (the ShopBot program) in Move/Cut mode (`SM`); the correct settings are already loaded, do not reload them → Spindle RPM window → key ENGAGED → dust collector on, at the collector.	`Reset` `SM`
	4	**Warm up, home** `C5`, the standard warm-up, if the spindle is cold. `C3` homes on the prox switches.	`C5` `C3`
SET UP	5	**Stock on** Spindle stopped, key OFF and removed, keypad closed, no file loaded or paused. Sheet flat, screwed where CAM said. Key back in, ENGAGED.	`key OFF`
	6	**Bit** Spindle stopped, key OFF and removed, keypad closed, no file loaded or paused. ER25 collet snapped into the nut, bit in (≥ 75 % of the shank inside), then thread the assembly onto the spindle; torque. Key back in, ENGAGED.	`key OFF`
	7	**XY zero** Jog to the file's XY origin (`K`): centre in the zeroing example, or corner if CAM uses a corner. Close keypad, `Z2`. Record the base coordinates.	`K` `Z2`
	8	**Z zero** Clip stays on the carriage plate. Plate on the right surface, input 1 test, `C2` (always), plate back in its holder.	`C2`

CUT

9 Air cut

`FP` with Offset = 3D, bit raised more than the deepest cut, spindle off. Then No Offset.

`FP` `Offset 3D`

10 Cut

RPM + Enter → `FP` → screen prompt → pendant START → spindle at commanded speed → OK → cut. Hand on the spacebar. Stay at the machine until it finishes.

`TR` `Enter` `FP` `START` `OK`

FINISH

11 Done

Finish or stop the file and stop the spindle. With everyone clear, raise Z and jog clear. Close the keypad, end any loaded/paused file and remove the key before cleanup. Collector off; clear, vacuum, brush and stow the bit. Report faults. Leave the machine powered for lab staff; users never power down. E-stop is for emergencies.

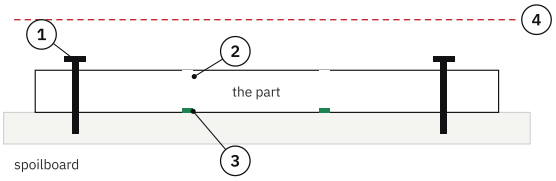
`Stop file` `key out`

TWO DRAWINGS

Hold-down and the collet, in section

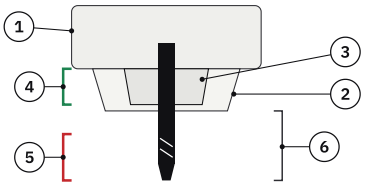
Get these two right and most first-job crashes do not happen.

HOLD-DOWN, IN SECTION



1. Screw in the waste.
2. Profile cut, 0.2 mm into the board.
3. Tab or onion skin (0.25–0.75 mm) keeps the part from flying.
4. Clearance height **above** every screw head and clamp.

COLLET, NUT AND BIT, IN SECTION



1. Spindle nose.
2. Standard ER25 nut supplied for the HSD AT/MT 1073-140. Snap the collet into the nut **first**, insert the bit, then thread the assembly onto the spindle. Use the tightening method demonstrated during induction; do not use a torque for a different nut.
3. Collet: clean, right size for the shank.
4. Shank ≥ 75 % in the collet.
5. Flutes never inside the nut.
6. Stick-out as short as the job allows.

Home, job zero and Z zero

Three different zeros. Mixing them up is the most common first-week crash.

C3
Machine home

Where the prox switches say 0,0. Fixed. Everything else is measured from here.

Z2
Job zero (X, Y)

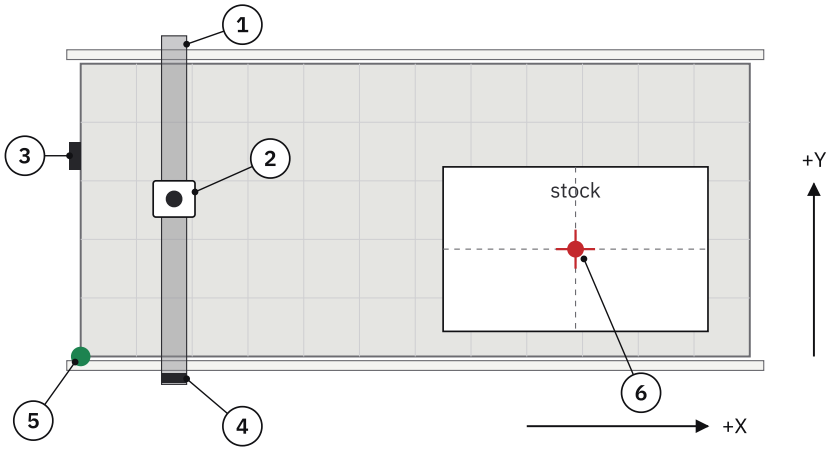
The point your CAM file calls X0 Y0. This example uses the centre of the material. Z2 zeros X and Y, not Z.

C2
Z zero

Material top *or* table surface — whichever `&PWZorigin` in the file says. Found with the plate.

FIGURE 1 · PLAN

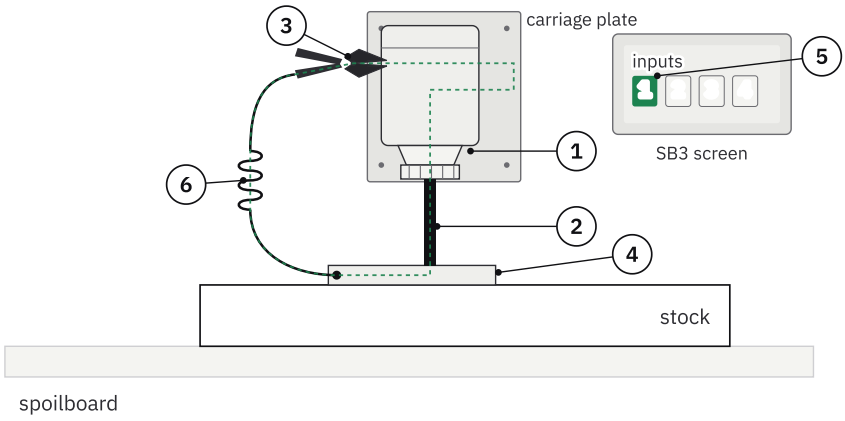
Home and job zero, from above



1. **Gantry** (X bridge): runs along the table on two X motors.
2. **Spindle on the Y car**: runs across the gantry.
3. **X proximity switch** · input 2.
4. **Y proximity switch** · input 3.
5. **Machine home 0,0.** `C3` finds it on the prox switches. Fixed.
6. **XY job zero — `Z2` at the material centre.** Match the centre origin in CAM. A corner origin is also possible, but only when the file uses it. Z2 does not set Z height; use C2 for Z.

FIGURE 2 · SIDE

Z zero with the plate



1. **Spindle and collet nut** on the aluminium carriage plate.
2. **Bit**: `C2` lowers it until it touches the plate, twice, then lifts.
3. **Ground clip**: it stays clamped on the aluminium carriage plate, on the edge. Leave it there; never on the bit or nut.
4. **Z-zero plate** on the surface the file expects (stock top here).
5. **Input 1** on the SB3 screen lights on contact, goes out when lifted. The green dashes are the circuit: plate → bit → spindle → carriage plate → clip → lead → plate.
6. **One coiled lead**, plate to clip. Nothing else to connect.

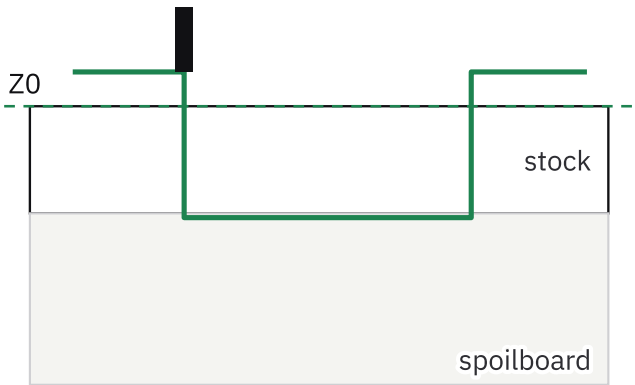
FIGURE 3 · THE CLASSIC CRASH

Same file, wrong surface

CAM assumed the **stock top**; you touched the plate on the **bed**. Every rapid and retract now runs *inside* the material and every cut goes a sheet thickness into the spoilboard: clamps hit, bit snaps. The reverse mismatch cuts air. Read `&PWZorigin` in the file header before C2, every time. [4,54]

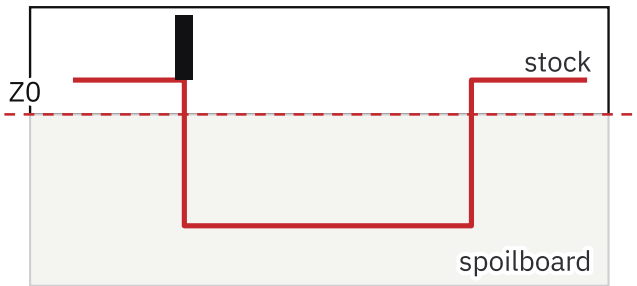
you touched the stock top

cuts as planned



you touched the bed

every move a sheet too deep



Match
File and plate both on the stock top: the rapid clears the sheet, the cut ends just below it.

Mismatch
File on the stock top, plate on the bed: the rapid runs inside the sheet, the cut goes a sheet thickness into the spoilboard.

Pre-flight checklist

TEAR-OFF SHEET · KEEP IT AT THE MACHINE

Print this section on its own and keep it at the machine. Tick every line, in order, every time. If a line cannot be ticked, stop and ask.

Operator _____ Second person _____ Date _____ Job / file _____

1 Before power

- [ ] Signed-off operator present; second person knows the E-stop (wall pendant by the monitor)
- [ ] Glasses, ear protection; no gloves, loose sleeves, hair, lanyards
- [ ] Table and rails clear of tools, offcuts, clamps from the last job
- [ ] Dust hose on the foot, brush skirt intact
- [ ] Key OFF and in your pocket for the bit change; bit correct and sharp, collet clean, nut torqued, stick-out minimal
- [ ] Stock: thickness measured and written down, flat on the deck
- [ ] Hold-down planned: screws/clamps outside every toolpath, tabs or onion skin for freed parts

2 Power and zeroing

- [ ] Control box on → E-stop twisted to release, pendant **Reset** pressed (input 4 not flashing) → PC on, SB3 open in Move/Cut mode (SM)
- [ ] Spindle RPM Control window open; interlock key in and ENGAGED; dust collector switched on at the collector
- [ ] C5 warm-up at the start of the day or after 4 h idle
- [ ] C3: machine homes on both proximity switches without complaint
- [ ] Read the .sbp header: mm, and which surface is Z0 (`&PWZorigin`: material top or table), feed, rpm, first Z move sane
- [ ] Jog to the CAM file's XY origin (centre or corner); Z2 (or ZX/ZY). The drawing uses the material centre
- [ ] C2: clip in place on the carriage plate, plate under the bit on the surface the file expects; touch plate to bit — **input 1 lights and goes out**; run C2 (two touches, lift); clip stays on the plate, Z-zero plate back in its holder on the car
- [ ] Keypad closed before FP

3 Running

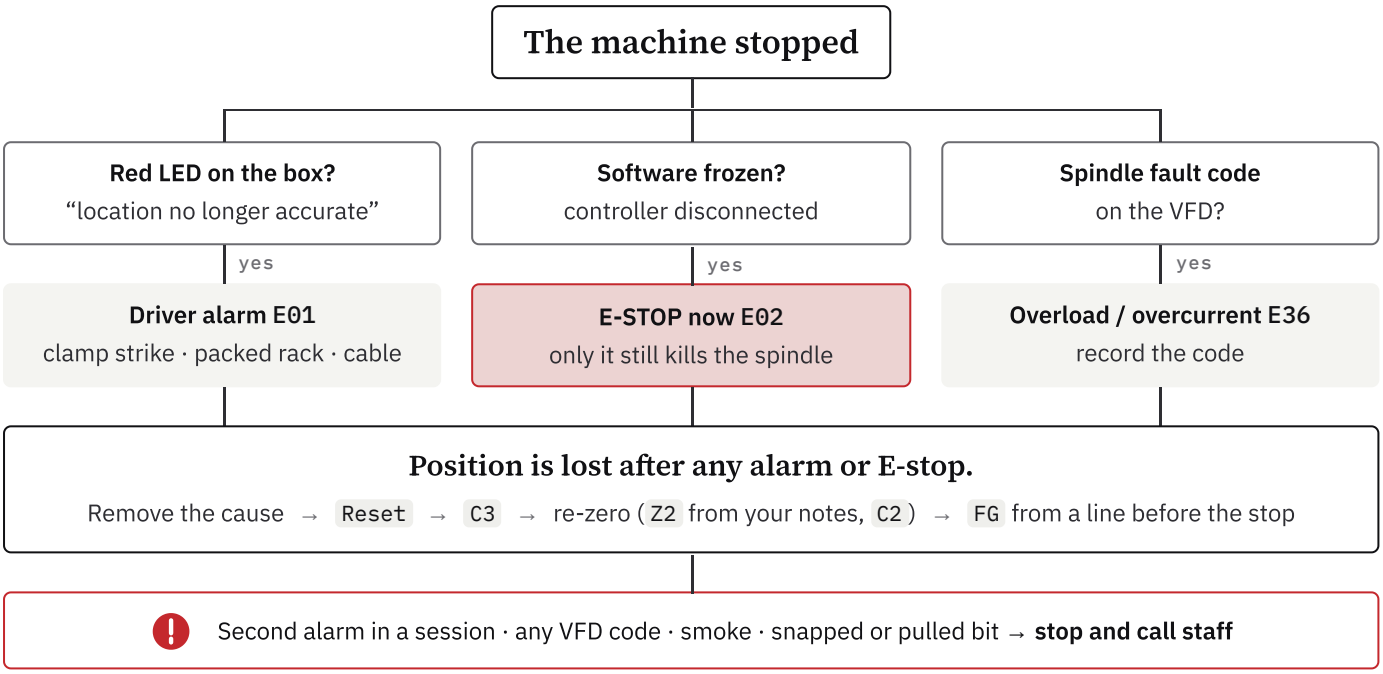
- [ ] Air cut a new file: FP with Offset = **3D**, bit raised above the stock by **more than the file's deepest cut**, spindle may stay off; then Offset = **No Offset** and C2 again if you moved Z
- [ ] RPM typed in the Spindle RPM Control window and **Enter**
- [ ] FP → prompt "Press START on control before pressing OK": pendant **START**, spindle at commanded speed, VFD shows the rpm, **then** OK (no separate VFD RUN)
- [ ] Hand near the spacebar, E-stop in reach, for the first plunge and the first pass
- [ ] Watch the chips: ribbons/curls good, dust and smoke bad
- [ ] Never leave the room while it runs

4 After

- [ ] File finished/stopped, spindle stopped; raise Z and jog clear with everyone outside the cutting area; collector off
- [ ] Key OFF and removed, keypad closed, no loaded/paused file before clearing parts, screws and offcuts; deck vacuumed, racks and rails brushed
- [ ] Bit removed, cleaned, back in its place; collet checked
- [ ] Material, cutter, feed, rpm and any fault recorded in your job notes; problems told to staff
- [ ] Machine left powered for staff. Users never power down; E-stop is for emergencies

Something went wrong

Start from what you see. Stop first, diagnose second.



Stop and call staff when

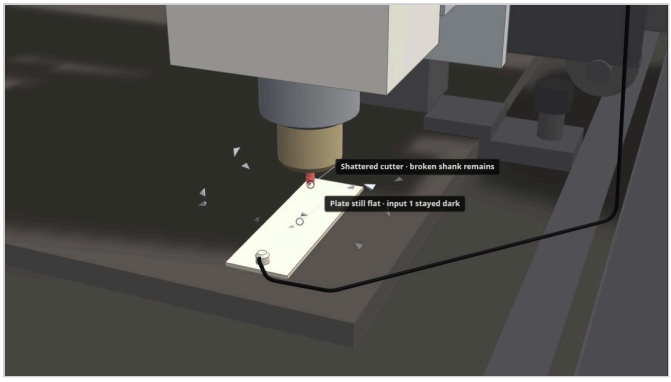
Stop (E-stop in an emergency). Before reaching in: spindle off and stopped, key OFF and removed, keypad closed, no file loaded or paused. Get a signed-off staff member before continuing if you see any of these:

- A driver alarm (red LED, "location no longer accurate") for the second time in a session.
- Any fault code on the spindle controller's display.
- C3 nags "move off prox" or runs into the end of the table.
- Input 4 keeps flashing after Reset; input 1 does not light in the circuit test.
- A damaged, pinched or hot motor or spindle cable; the dust hose off the foot or split.
- Smoke, embers, burning smell, or hot dust under the brush.
- A bit that pulled out, snapped, or hit a screw or clamp.
- Anything in the glossary marked **staff only** (VN, VL, SF, ZT, TS offsets, eccentrics, pinion re-engagement, VFD parameters).

SEEN IN THE 3D MODEL

Five faults, frozen

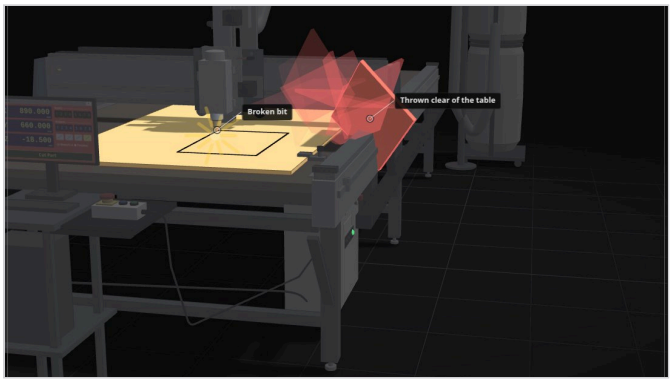
Stills from the site's 3D model, which acts out ten faults. The E-number leads to the row in Appendix A.



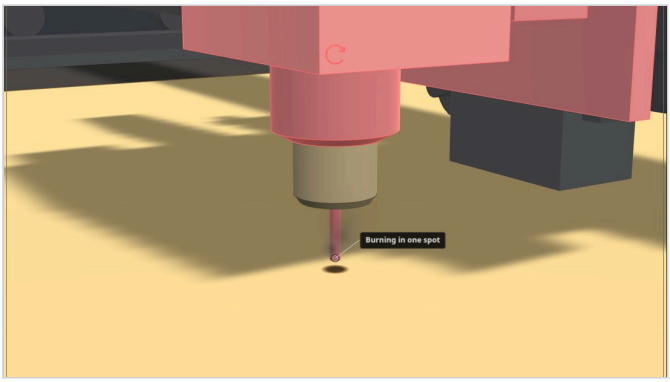
E05 · C2 does not see the plate. Input 1 stays dark, Z keeps driving down and the cutter shatters on the plate. Test input 1 before every C2.



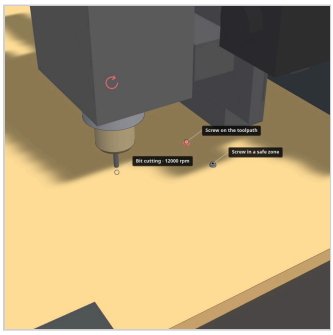
E10 · Bit slips in the collet. 2.5 mm of slip turns an 18 mm cut into 20.5 mm. The motors cannot see it.



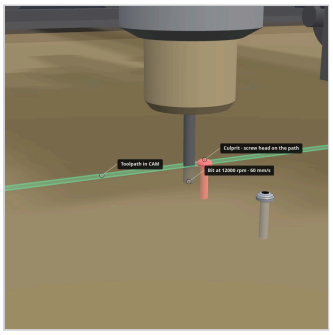
E22 · Part lifts off. No tabs or onion skin: the freed part is thrown clear of the table and the bit snaps.



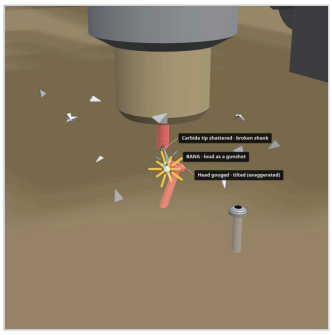
E02 · USB lost. Motion stops but the spindle keeps turning and burns in one spot. Only the E-stop kills it.



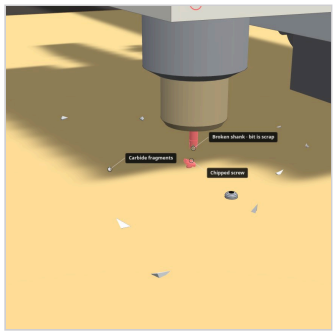
1 Screw on the toolpath, bit cutting at 12 000 rpm.



2 The CAM toolpath runs over the screw head.



3 Impact: carbide tip shatters, a bang like a gunshot.



4 Aftermath: broken shank, fragments, a chipped screw.

E39 · Bit hits a hold-down screw, in four beats (key S in the online 3D model). Check every screw and clamp against the toolpaths in CAM before you run.

Start from what you see

Find what you see, not what you suspect. Each card names the most likely causes in order and points to the fault entry (E-number) in Appendix A.

Everything stopped dead, red LED on the box, "location no longer accurate"

Alpha driver alarm. Stop and call staff; do not open the cabinet. Staff check the driver alarm and cause (clamp strike, packed rack, cable). After recovery, **rehome and re-zero**. → E01

Software froze or says the controller disconnected; spindle still running

USB dropped: the PC went to sleep or the cable was knocked. **Press an E-stop**: the software can no longer stop the machine and the E-stop is the only control that kills the spindle. Then fix the cause, Reset, C3, C2, resume with FG. → E02, E03

C2 drives the bit into the plate and keeps going

Input 1 not seeing the touch: clip off, dirty plate, open lead. **Spacebar**; if Z still moves, **E-stop**. The circuit test (input 1 lights on contact) before C2 is the only guard against this. → E05

C2 touches once and then stops with an error

Plate moved before the second touch; or (staff: check the C2 settings) a zero Safe-Z pull-up. → E06

C3 runs into the end of the table, or nags "move off prox"

Proximity switch gap, polarity, or input channel. Watch inputs 2 and 3 light while you hold a screwdriver on the switch. → E04

Spindle did not start but the axes move

Interlock key, output 1, pendant START not pressed before OK, VFD in LOCAL. Spacebar before the bit reaches the stock; E-stop if it already has. → E07, E09

Typed a new RPM, nothing changed

Enter not pressed in the Spindle RPM Control window, window closed, or RPM controller not talking to the VFD. Read the rpm on the spindle controller's display, not in the software. → E08

Cut gets deeper every pass; the last pass goes through the spoilboard

Bit slipping in the collet. The motors cannot see it. Stop, re-torque or replace the collet, re-zero Z on an uncut area. → E10, E11

Snapped bit

Was the spindle turning? Then: chipload too high or too low, straight plunge too fast, dull cutter, or it hit a clamp/screw. → E12, E21, E39

Loud bang, bit broke mid-cut

With the spindle running, the bit hit metal: a hold-down screw or a clamp on the path, or a clearance height below a screw head. **Spacebar**, spindle off, key OFF, tell staff; the bit and possibly the collet are scrap. Before the next run, check every screw and clamp position against the toolpaths in CAM. → E39, E21, E28

Melted plastic, aluminium welded to the flutes

Feed too slow for the rpm, chips being recut. Faster feed, shallower depth, single flute for plastics. → E13

Frayed plywood veneer, fuzzy edges

Up-cut tooling on the top face, dull bit, wrong direction. Compression or down-cut bit, climb finishing pass. → E14, E24

Burn marks, smoke, hot dust

Rubbing, not cutting: too slow, dull. Stop and check the dust under the brush for embers. → E15, E25

Second pass offset from the first; circles come out oval; sides not square

Position creep: packed rack teeth, worn pinion, loose pinion set screw, tired motor spring, or roller wheels wrong. → E16, E17, E18, E19, E38

Ticking or a bump at the same place every time

Debris on the V-rail or in the rack at that spot. Clean it; do not grease over it. → E16, E20

Part lifted, moved or flew; sheet wandered

Hold-down failed as the profile cut through: no tabs/onion skin, bowed stock, too few screws. → E22, E23

Right shape, wrong place on the sheet

Job zero not where CAM thought: wrong corner, wrong Z surface (`&PWZorigin` in the file header vs the surface you touched), zeroed with Z2 before homing. Air-cut first next time. → E26, E32

Everything is 25.4× too big or too small; feeds absurd

Inch/mm mismatch between the file and the software (VD). → E27

Rapid move hit a clamp

CAM clearance height below the clamp. → E28, E21

"Z limit" or "table limit" message

Retract height beyond the 6/8 in Z stroke, or the job runs past the table limits (VL). Never disable limits to get there. → E29, E35

Preview plays, nothing moves

Software in Preview (SP) instead of Move/Cut (SM), or no controller connection. → E33

An axis moved when I touched the keyboard

The keypad window still had focus. Close it when you are done jogging. → E34

Spindle stopped with a code on its controller display

Overcurrent/overload. Stop the file, record the code, find the cause before resetting. → E36, E37

I stopped the job and need to continue

Decide whether position survived (spacebar stop: yes; E-stop or driver fault: no → Reset, C3, re-zero). Then restart from a line with FG, spindle started by the start sequence. → E30, E31

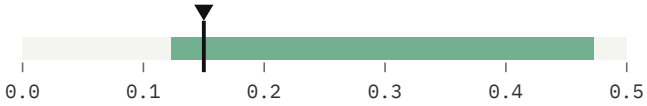
Feeds, speeds and the post

One number decides most cuts. Most machine failures are decided at the computer.

Spindle speed	14000 rpm
Feed	4200 mm/min
Flutes	2
Cutter Ø	6 mm

CHIPLOAD F_Z

0.150 mm per tooth



In range. Plunge at 2100 mm/min or less.

$f_z = feed \div (rpm \times flutes)$

Green band: ShopBot's plywood range for a ¼ in cutter (0.13–0.5 mm/tooth), scaled to the cutter Ø. Too thin: rubbing, heat, burn. Too thick: deflection, chatter, snapped bit. Plunge at ≤ half the feed.

The site's chipload calculator, frozen at the worked example: 6 mm two-flute, 14 000 rpm, 4200 mm/min.

The cutter removes a chip every time a flute passes. The thickness of that chip, the **chipload** f_z, decides almost everything: too thin and the edge rubs, heats, dulls and burns; too thick and the cutter deflects, chatters and snaps.

$$f_z = feed \div (rpm \times flutes)$$

Worked example (the calculator's default): 4200 mm/min ÷ (14 000 rpm × 2 flutes) = **0.15 mm per tooth**, a reasonable chip for a 6 mm cutter in plywood. **CAM works in mm/min; SB3 and the .sbp file use mm/s — divide by 60** (4200 mm/min = 70 mm/s). [11]

Rules of thumb. Plunge at half the XY feed or less; ramp into pockets. Keep the bit's stick-out as short as the job allows. If depth of cut is 2× the diameter, reduce chipload 25 %; at 3× diameter, 50 %. A spindle loses torque at low rpm: if it bogs, raise rpm and feed together to keep the chip the same. Record the material, cutter, feed, rpm and any fault in your job notes and report problems to staff. [11,59]

STARTING POINTS

Used in Icelandic Fab Labs

6 mm two-flute unless stated. Record the material, cutter, feed, rpm and any fault in your job notes and report problems to staff.

MATERIAL	CUTTER	RPM	FEED MM/MIN	MM/TOOTH	NOTES
Birch / spruce plywood	6 mm 2-flute	14 000	4 200	0.15	Labs report 2000–6500 mm/min (0.07–0.23 mm/tooth); below ~3600 mm/min at 14 000 rpm you are under ShopBot's range. 3 mm pass, 40 % stepover; down-cut to 3 mm then up-cut to 0.2 mm past the thickness; tabs 12 × 3 mm. To go gentler, drop rpm, not feed. [11,56,57,58]

MATERIAL	CUTTER	RPM	FEED MM/MIN	MM/TOOTH	NOTES
MDF	¼ in	—	similar to plywood	0.33–0.40	Fine dust: extraction and a mask. MIT's table; plunge ≤ 50 % of the XY feed. [59]
EB700 epoxy tooling board	12 mm 3-flute	8 000	8 400	0.35	Plunge 300 mm/min; adaptive 20 mm max / 5 mm fine stepdown. One worked job, not a validated recipe; respirator. [61]
PU foam / tooling foam	—	—	—	0.28–0.33	Cuts easily; the dust is the hazard. [59]
Plastics (HDPE, acrylic)	1-flute (O-flute)	low	high	—	The test is a clean curl, not a weld. [11,35]
Aluminium	1-flute	—	high	—	Marginal on a router: shallow depth, lubricant; if it welds, less depth and more feed. [35]

CHECKLIST

From CAM to the machine

Most "machine" failures are decided at the computer. Go through this list every time you post a file. Post with Autodesk Fusion's official ShopBot (OpenSBP) post processor. The correct machine settings are already loaded on the control PC; setup is then checked by hand at the machine. [54,63]

1. **Units.** File and software agree (mm). If SB3 stops with UNIT_ERROR, the file and SB3 disagree on mm/inch: re-post the file in mm. Do not change or reload the machine settings — they are already correct on the control PC; ask staff if the units still disagree. [10,54,63]
2. **Origin corner and Z reference.** Which corner is X0 Y0? Is Z0 the **material top** or the **table surface** (`&PWZorigin` in the posted file)? It must match what you touch with C2. Bed zero with a top-zero file puts every rapid and retract *inside* the stock and every cut a sheet thickness into the spoilboard; the reverse cuts air. [4,6,54]
3. **Stock thickness measured**, not assumed. Plywood sold as 18 mm is 17.4–18.6 mm; through-cuts and onion skins depend on it. [56]
4. **Clearance / retract height** above the tallest clamp or screw head, with margin — and below the Z stroke. [10,12]
5. **Hold-down planned in the CAM**: screw positions in the waste, screws and clamps clear of every toolpath including the tool's radius; tabs or a 0.25–0.75 mm onion skin for every part that will be freed. Plan the full cutter envelope in CAM and check it in the simulation. [12,63]
6. **Feeds, rpm, flutes, depth per pass** computed, not copied; check the first-pass depth in the simulation and posted file before running. [11,62]
7. **Simulate** and look for: straight plunges into solid stock, rapids posted as feed moves above the stock (slow and dangerous at clamp height), curves turned into thousands of tiny steps (the machine stutters) — ask staff about the post settings. [62]
8. **One file per tool** unless you are sure of the tool-change prompts; name files with the tool and the date. [62]
9. **Open the .sbp in a text editor** and read the header: units, `MS` feed, `TR` rpm, `&PWZorigin`, the first `JZ` and `M3`. If a number surprises you, it will surprise the machine. [10,62]

Looking after the machine

The published schedule and why it matters here. Glossary, I/O table and sources are in the appendix.

I Every session (you)

- C5 warm-up at the start of the day
- Vacuum deck, brush racks and rails
- Check the brush skirt and that the hose is on the foot
- Listen: ticks and bumps are debris
- Record material, cutter, feed, rpm and faults in your job notes; report problems

II Every 40 h (staff)

- Wipe rails, light grease on the racks (BB-sized dab per 6 in, work in, wipe excess)
- Check pinion set screws and motor spring tension
- Check collet, nut and spindle nose for scoring

III Quarterly (staff)

- Roller/eccentric preload on the car and Z
- Rack and pinion wear, re-engagement
- Cable carriers and connectors
- VFD fan runs

IV Semiannual (staff)

- Check gantry square (diagonal test), table flatness
- Spoilboard surfacing
- Proximity switch gap and function
- Replace collets at 400–700 h or any incident

WHY

Why maintenance matters here

Students clean accessible surfaces, inspect tooling and report faults. Mechanical adjustments, lubrication and servicing are staff tasks.

- **Daily:** C5 warm-up and listen for the fan; inspect bit, collet, nut, hose; check alignment without pushing the gantry; tell staff if it is out of square. Jog, C3, C2. [13,4]
- **Every 40 h:** brush X/Y/Z racks and pinions, then a BB-sized blob of light grease every 6 in, run the axes ~10 times, wipe the excess; Scotchbrite + WD-40 on the V-rails, brass brush on the rollers; push-pull play check at every drive. [13]
- **Quarterly / semiannually:** spin-test rollers, tighten pinion set screws, check Z eccentrics and spindle square; back up settings with US (restore with UR). [13,18]
- **Semiannually at 40 h/week:** all pinions replaced, Z extrusion stripped, cleaned and re-set, collets replaced as required. [13]

On lubrication. The received wisdom to run racks dry is not ShopBot's instruction. Their 2016 schedule says light grease; the 2014 troubleshooting guide says heavy grease; the 2022 assembly guide uses lithium grease on the pinions. The sources disagree on formulation, not on greasing — and all of them say clean first, never grease over packed chips. [2,13,15]

Self-check

Twelve questions. Passing is part of the sign-off.

Passing this is part of the sign-off. Answers are under each question; think before you open them.

1 The file is loaded, you clicked OK on the spindle prompt and the gantry is moving toward the stock — but the spindle is silent. What do you do in the next two seconds, and what went wrong?

> ANSWER · Spacebar now; E-stop if the bit is already in the stock. You clicked OK before pressing START on the pendant (or the key/output 1 was off). See the start sequence and E07, E09.

2 You press the E-stop halfway through a job. Which of X, Y and Z zero can you trust afterwards, and what must you do before restarting?

> ANSWER · None. Release, pendant Reset, C3 to rehome, jog to the recorded job-origin coordinates and Z2 again, C2 for Z, then FG from a line before the stop. See Stops, E30.

3 C2 ran and reported Z zero, but the first pass cuts 2 mm too deep across the whole sheet. Name three possible causes and the test for each.

> ANSWER · Wrong plate thickness in the settings (staff measure the plate and check TS); zeroed on the wrong surface for the file's `&PWZorigin` (read the header); bit slipping in the collet (witness mark on the shank, staff check after correct tightening). E05, E10, E32.

4 The second profile pass is 1 mm to the left of the first. The alpha drivers did not alarm. Why could they not have noticed, and where do you look?

> ANSWER · The loop is closed around the motor shaft: a pinion that jumps teeth or slips on its shaft, or packed chips in the rack, move the gantry without the motor knowing. Look at rack teeth, pinion set screws, motor spring tension, roller wheels. E16–E19.

5 Your CAM file has Z0 at the material top; the lab convention for through-cuts is Z0 on the bed. What happens if you zero on the bed and run it anyway?

> ANSWER · Every rapid and retract runs inside the stock and every cut goes a full sheet thickness into the spoilboard: crashed clamps, broken bit. Match `&PWZorigin` to the surface you touch, every time. E32.

6 You want 0.15 mm/tooth with a 6 mm two-flute at 14 000 rpm. What feed in mm/min? What plunge?

> ANSWER · 14 000 × 2 × 0.15 = 4200 mm/min; plunge at half or less, ≤ 2100 mm/min, ramped if possible. Feeds.

7 You pressed the spacebar to pause and want to check the cut. What must you confirm before your hand goes near the bit?

> ANSWER · A spacebar stop is not isolation. Before reaching near the bit, confirm that the spindle has stopped, the interlock key is OFF and removed, the keypad is closed and no file is loaded or paused. Keep clear before re-engaging the key or restarting. Stops.

8 The part you are cutting out is 300 × 200 mm. List three ways to keep it from flying on the last pass, and the drawback of each.

> ANSWER · Tabs (hand finishing); onion skin 0.25–0.75 mm (hand finishing, needs accurate thickness); screws through the part into the spoilboard (holes, must be clear of toolpaths). E22, E23.

9 What does the alphaStep driver's closed loop see, and what does it not see? Give one failure from each category.

> ANSWER · Sees rotor position vs command: a stalled motor from a clamp strike alarms (E01). Does not see anything downstream of the shaft: a slipping pinion, a creeping bit, a moving sheet (E17, E10, E22).

10 After the job: name five things you do before you walk away.

> ANSWER · Finish or stop the file and stop the spindle; raise Z and jog clear with everyone outside the cutting area; close the keypad, end the loaded/paused file and remove the key; clear and clean the deck and stow the bit; report faults. Leave the machine powered for lab staff. Users never power down the machine; E-stop is for emergencies.

11 Name the three buttons on our pendant and what each does.

> ANSWER · Reset re-enables the drives after an E-stop (twist the E-stop to release it first); green Start starts the spindle (when output 1 and the key allow it); E-stop cuts spindle power at once and motor power about 5 s later. Stops.

12 Where does the Z-plate's clip go, and where does the plate go back to?

> ANSWER · Nowhere new: the clip stays clamped on the aluminium carriage plate; the circuit runs through plate → spindle → collet → bit. The plate goes back in its holder on the YZ car with the lead coiled. Never put the clip on the bit or nut: the spindle whips it off.

Reference

The full fault table, the command glossary and the sources. Look things up here; learn them from the chapters.

A All 39 failure modes

What you see, why, what to do. Faults the 3D model can act out carry their key. Numbers in brackets are sources (Appendix C).

E01
Alpha driver fault — “location no longer ACCURATE”
[1,22,30] 3D key 1

SEE Abrupt stop, driver-fault dialog, red LED on the box; coordinates untrustworthy.
WHY Obstruction or excessive load, chips in the rack, a struck clamp, a damaged motor cable; AS drivers also trip on heat, voltage, speed and sensor faults.
DO Stop and call staff. Staff check the driver LED and cause; students do not open the cabinet. After recovery, reset, re-home (C3) and re-zero. Reduce feed or depth if it was the cut.

E02 USB disconnect / communications failure [7,55] 3D key 8

SEE SB3 loses the controller mid-file; spindle keeps running.
WHY PC power management (USB selective suspend, sleep), long or damaged cable.
DO Dedicated PC with sleep and selective suspend off, direct cable under 10 ft, no extension; rehome afterwards.

E03 Dust hose off the foot / skirt torn [13] 3D key 8

SEE Chips everywhere, dust in the air, the brush not sweeping.
WHY Hose pulled off the foot by a long move; brush skirt torn or lost.
DO Stop, spindle off, refit the hose or tell staff; never cut without the foot on.

E04 C3 homes early, fails, or asks to “move off prox”
[2,6,49] 3D key 3

SEE Wrong zero, stop before the target, or a nagging prox dialog.
WHY Misaligned target, swapped input polarity, bad switch or cable, dead input channel.
DO Trip the switch with a screwdriver and watch inputs 2/3; fix alignment or the fault, don't disable limits.

E05 Z plate not detected [4,6] 3D key 2

SEE Input 1 stays dark; the bit keeps descending.
WHY Clip knocked off the carriage plate or on paint/dust, a dirty plate or bit, dirty contact, open lead or connector.
DO Plate-to-bit test with clip on before every C2; watch input 1 light and clear.

E06 C2 touches but stops [4,6] 3D key 2

SEE First touch registers, routine halts.
WHY Plate moved before the second touch; or (staff: check the C2 settings) a zero Safe-Z pull-up.
DO Positive pull-up in the active units; leave the plate until both touches finish.

E07 Spindle will not start [4,21,24,25] 3D key 4

SEE Axes move; spindle stays still; a still bit snaps instantly.
WHY Interlock key off, no output 1 / pendant Start, VFD in LOCAL, or a drive fault.
DO Key on, output 1, Start; check the VFD command source; listen before the bit enters stock.

E08 RPM will not change [4,6,52] 3D key 4

SEE Typed value ignored; speed stays as before.
WHY RPM window closed, Enter not pressed, wrong VFD selected, RPM-board USB lost, legacy keypad-only VFD.
DO Open TR, pick the fitted drive, type, Enter; confirm on the VFD display.

E09 Wrong spindle-start order [4] 3D key 4

SEE File starts with a still spindle.
WHY Pendant green START pressed after OK (alpha wants it before), or not at all.
DO At the spindle prompt: pendant START, hear it reach speed, read the rpm on the VFD, then OK.

E10 Bit slips in the collet [4,13,16,31] 3D key 5

SEE Cuts get deeper pass by pass; Z zero seems to drift.
WHY Dirty or worn ER25 collet, nut not snapped onto the collet first, wrong torque, wrong shank size.
DO Clean taper/collet/nut; collet into nut first; use the tightening method demonstrated for the standard ER25 nut supplied for the HSD AT/MT 1073-140, not a torque for another nut; replace collets after 400–700 h or any incident.

E11 Tool pull-out read as axis drift [1,13,16] 3D key 5

SEE Depth changes while coordinates look right; witness mark on the shank has moved.
WHY Inadequate grip; the motor encoder cannot see a bit sliding.
DO Compare Z zero before/after a test cut; replace suspect collet and bit; don't clamp on the flutes.

E12 Broken bits — chipload, plunge, ramping [10,11,25,26]

SEE Snapped cutter, chatter, spindle bogging.
WHY Wrong feed/rpm/depth, aggressive straight plunge, dull cutter — or a spindle that was never running.
DO Maker's chipload chart, ramped entries, reduce depth; distinguish VR acceleration ramping from CAM entry ramps.

E13 Plastic melts / aluminium welds to the cutter [11,35]

SEE Gummed flutes, fused chips, repeated breakage.
WHY Feed too low for rpm, recutting chips, heat.
DO Make chips not heat: faster feed, shallower depth, proper evacuation.

E14 Plywood tear-out [11]

SEE Frayed veneer, rough edges.
WHY Cut direction, dull or wrong geometry, grain, feed.
DO Sharp compression or down-cut tooling, test both directions on scrap; don't slow down until it burns.

E15 Burning in hardwood [4,11]

SEE Scorched edge, smoke, hot dust.
WHY Too slow, rubbing with a dull tool.
DO Sharp tool, correct chipload; stop and check for embers; never leave a running job.

E16 Dust packed in rack teeth [13,32] 3D key 7

SEE Ticking, uneven motion, erratic Z.
WHY Chips filling the gear mesh.
DO Brush racks and pinions clean, then light grease (schedule: BB-sized blob every 6 in, run the axes, wipe excess).

E17 Worn rack or pinion [13,15,16,33] 3D key 7

SEE Oval circles, Z notches at certain heights, reversal play.
WHY Worn teeth or poor mesh.
DO Push-pull test, inspect full travel, replace pinions as a set; CC circle test separates mechanics from bad CAD.

E18 Loose pinion set screw / tired motor spring [2,13,15] 3D key 7

SEE Clunks, sudden registration loss, backlash.
WHY Pinion moving on its shaft or not fully engaged.
DO Tighten set screw and key, 3.5 turns past contact on spring mounts, full-width engagement; rehome.

E19 Roller bearings loose or over-tight [13,16,18,19]

SEE Lateral Z slop, bad fourth pocket wall, binding, axis faults.
WHY Eccentrics wrongly set, contamination, worn bearings.
DO Adjust eccentrics incrementally with drives disengaged; replace rather than overtighten.

E20 Debris on the V-rails [13,16]

SEE Bumps that repeat at one spot.
WHY Build-up on rail edges or rollers.
DO Scotchbrite + WD-40 on rails, brass brush on rollers, every 40 h.

E21 Crash into a clamp or screw [1,4,12] 3D key 9

SEE Bang, broken cutter, alarm, shifted cut.
WHY Toolpath or rapid intersects hardware.
DO Stop, inspect tool/mount/drives, fix clearance, rehome and re-zero; plan screw zones.

E22 Part lifts or flies [4,12,29] 3D key 6

SEE Sheet rises or a cut-out catches the bit and is thrown.
WHY Poor hold-down, bowed stock, no tabs.
DO Screws in the waste, tabs or 0.25–0.75 mm onion skin, never reach in.

E23 Clamps or shims let go [12,11] 3D key 6

SEE Sheet shifts after the first passes; parts come out the right shape in the wrong place.
WHY Spring or pistol-grip clamps creeping, shims loosening as the sheet relaxes, too few screws, screws only at the edge.
DO Screws into the spoilboard through the waste, at least every 300–400 mm and near every freed part; re-check clamps after the first pass; shim bowed stock flat before screwing.

E24 Climb vs conventional [11,12,16]

SEE One direction chatters, the other is clean.
WHY Edge engagement changes with direction; looseness compounds it.
DO Verify mechanics and hold-down first; climb on the finishing pass often helps, test on scrap.

E25 Fire in the dust path [4,36,43]

SEE Sparks, smoke, smouldering dust.
WHY Friction heat, screws hitting the impeller, hidden fasteners.
DO Never unattended; keep fasteners out of the intake; extinguisher within reach.

E26 Wrong XY zero / origin corner [4,26,42,47]

SEE Right shape, wrong place.
WHY Job origin differs from the CAM origin; jogging confused with zeroing; Z2 and C3 mixed up.
DO Home with C3, move to the chosen origin, Z2; verify the CAM corner; air-cut first.

E27 Inch/mm or SA/SR mismatch [10,11,27,54]

SEE Toolpath scaled or offset; speeds absurd.
WHY Active units don't match the file; wrong distance mode; edited unit values.
DO Check units in Values → Display before loading; the Autodesk post's CN,90 header only checks units, it doesn't set them.

E28 Retract height too low for clamps [10,12] 3D key 9

SEE Traverse hits a clamp though the contours clear.
WHY CAM clearance below fixture height, or Z reference changed.
DO Clearance = tallest fixture + margin; preview rapid links.

E29 Retract beyond Z stroke [1,2,10]

SEE Z-limit message, top-stop approach.
WHY Safe height + offset + tool length exceeds usable Z travel. Have staff check that the setup fits before running.
DO Revise fixtures/toolpath; don't disable limits to reach the unreachable.

E30 "Stop button pressed" / won't resume [4,11]

SEE Stopped dialog, no motion.
WHY Input 4 still active, E-stop latched, driver not reset.
DO Clear the cause; after an electrical E-stop rehome and re-zero — positions did not survive.

E31 Restarting a broken job by line [10]

SEE Half-cut part; restarting from line 1 recuts or collides.
WHY Restart point lacks speed, spindle, offset state.
DO Re-establish zeros, FG (File Load in Goto mode), pick the line, approach at verified safe Z.

E32 Table-surface vs part-surface zero [4,13]

SEE Rapids and retracts run inside the stock, cuts go a sheet thickness into the spoilboard (bed zero, top-zero CAM) — or the whole job cuts air (the reverse).
WHY C2 zeroed to one surface, CAM assumed the other.
DO Match CAM zero option to the touched surface at every setup.

E33 Stuck in Preview [10,34]

SEE Animation, no motion.
WHY SP active instead of SM; or no controller connection.
DO Check connection, then SM; never toggle with hands in the envelope.

E34 Accidental keyboard jog [4,47]

SEE Axis moves when an arrow key is pressed.
WHY Keypad window has focus.
DO Exit keypad when done; Fixed increments for fine approach.

E35 Table limit trips mid-cut [2,10,11]

SEE Limit warning near an edge.
WHY Real limit, wrong base offset, VL/SF/VN settings, false input.
DO Check coordinates and switch state; limits use the base system, not Z2 zero.

E36 VFD overcurrent / overload [24,30,39]

SEE Spindle stops with a drive code; gantry may keep moving.
WHY Load, acceleration demand, short, loose motor wiring.
DO Stop, record model and code (Delta oc/oL vs Yaskawa oC/oL1/oL2 — different tables), fix the cause before resetting.

E37 Motor/driver overheating, spindle fan dead [2,13,22,41]

SEE AS code 1, thermal stop, no fan noise.
WHY Driver heatsink at ~85 °C, blocked cooling, failed 24 V fan.
DO Cool down, check vents and fan cable, match load to capacity.

E38 Gantry out of square [2,13,16] 3D key 1

SEE Diagonals differ; corners not 90°.
WHY Gantry reset with sides misaligned; unequal mesh; table shifted.
DO Stop and call staff for gantry alignment and diagonal checks; validate squareness before cutting again.

E39 Bit hits a hold-down screw [1,4,12] 3D key S

SEE A loud bang mid-cut with the spindle running; the bit breaks off and a screw head in the waste is chewed up.
WHY A screw placed on or too close to a toolpath, or the CAM clearance height set below the screw heads so a move between cuts runs into one. Carbide shatters on steel.
DO Spacebar, spindle off, key OFF, tell staff; the bit and possibly the collet are scrap. Screws only in planned safe zones of the waste, drawn in CAM and checked against every toolpath and rapid; clearance above the tallest screw head.

B Command glossary

MS	Set Move (feed) speed, XY/Z/A/B, in/s or mm/s	[CR 20]
JS	Set Jog (rapid) speed, XY/Z/A/B	[CR 16]
VS	Set all 8 move and jog speeds in one sheet	[CR 42]
TR	Opens Spindle RPM Control; `TR,n` requests n RPM in a file (SB3.5.5+)	[QS 15–16]
C2	Z-zero with the plate (input 1): two touches, then pull-up	[QS 14]

C3	XY homing using the proximity switches (inputs 2/3)	[QS 15; UG 20]
C5	Spindle warm-up: 9000/10000/12000 RPM for 3 min each, then off	[QS 16]
C6, C7, C9	Custom-cut slots. Student procedures use the standard C2, C3 and C5 routines. Do not remove C-commands from posted files	—
JZ	Jog Z to a location (or by a distance in relative mode)	[CR 17]
JH	Safe-Z, then jog to XY 0,0	[CR 17]
J2	Jog X,Y at Jog speed	[CR 15]
J3	Jog X,Y,Z at Jog speed	[CR 16]
M2 / M3	Move XY / XYZ at Move (cutting) speed	[CR 17–18]
MZ	Move Z at Move speed (`MZ,0` checks Z zero)	[CR 21; TM 11]
SA / SR	Absolute / relative distances; use absolute in part files	[CR 25; CR 28]
VD	Display values: linear units 0 = in, 1 = mm (3rd field), keypad fixed distance, logs	[CR 35; UG 21]
VL	Table limits in base coordinates, plus a limit-check flag	[CR 38; UG 20]
VN	Input assignment: limits on/off, input 4 stop mode, input modes	[CR 38–39]
VI	Comm port and driver channel settings (staff only)	[CR 37]
VO	Temporary tool offset (dual Z / air drill)	[CR 40]
SO	Set output n on/off: `SO,1,1` = output 1 on	[CR 27; UG 48]
SK (K)	Keypad arrow control; Move mode only	[CR 26]
ZZ	Zero Z here	[CR 45]
Z2	Zero X and Y here	[CR 43]
Z3	Zero X, Y and Z here	[CR 44]
ZX / ZY	Zero X only / Y only	[CR 44–45]
ZT	**Staff only.** Redefines the table base coordinates that limit checking and C3 rely on; never use it for a job origin (use Z2). ST resets the working display to base	[CR 44; CR 28]
FP	Load and run a part file (fill-in sheet: proportion, reps, offset, tabs)	[CR 12]
FG	Run a file in Goto/single-step mode; resume from the last line	[CR 11]
SP / SM	Preview mode / Move-Cut mode	[CR 28; CR 27]
PAUSE	In a file: wait for a key (`PAUSE n` = n seconds)	[UG 27]
CN,90	Calls custom cut 90; the CAM post uses it to expose controller variables for a units check	[54]
&var	Part-file variable, e.g. `&Zdepth = -0.52` then `MZ,&Zdepth`	[UG 28]
MO	Staff only: motors off and all outputs off. Not an end-of-job command	[CR 20]
TS	ShopBot Setup: plate thickness, prox-switch offsets	[CR 30; TM 12]
SF	File and move limit checking on/off	[CR 26]

Inputs, outputs and status LEDs

Input 1	Z-zero plate. Lights when bit and plate touch	[QS 14]
Input 2	X proximity switch, configured as a limit (VN)	[CR 39; UG 46]
Input 3	Y proximity switch, configured as a limit (VN)	[CR 39; UG 46]
Input 4	Stop circuit. On an alpha: normally-closed E-stop, wired ES1–ES4; cannot be reassigned	[CR 39; UG 47]
Input 5	Z proximity switch	[2]
Output 1	Spindle/router (on an alpha, used with the pendant Start)	[QS 12–13; UG 74–75]

Output 4	Safety/"tool active" output; on during motion and with the keypad open	[CR 27; UG 48; QS 12]

Reading the indicators

- An active input turns **green** in the input section of the Position window [UG 46].
- Output 4 lights when the keypad opens [QS 12]. Its default mode is −1: on with the warning signal before a move, off at the end of it [CR 27].
- Inputs are 5 V TTL. An input reads ON when pulled to ground, which is why the Z-plate clip closes input 1 [UG 46].
- Outputs are 5 V TTL, about 5 mA. They signal relays and must not power accessories directly [UG 47].
- **Motor driver LEDs (alpha AS-type drivers):** green **OPERATION**, red **ALARM**. The red LED blinks a fault code.

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Manual tags: AG Assembly Guide 2022 [2] · QS Quick Start 2016 [4] · UG User Guide 2015 [11] · CR Command Reference 2015 [10] · HD Holding Down [12] · MS Maintenance Schedule [13] · TM Troubleshooting Mechanicals [16] · XY Manually Setting XY Zero [47].